



DBT Training Curriculum (40 Hrs + 4 Hrs Assessment)

Prerequisites

- Working knowledge of SQL (joins, aggregations, CTEs).
- Basic Python knowledge (variables, functions).
- Familiarity with data warehousing (schemas, facts/dimensions).
- Basic Git usage (clone, commit, branch, push).

1. DBT Setup & Project Structure (Sessions 1–2, 4 Hours)

Session 1 – Installing and Configuring DBT

- Concept: Why DBT (vs. traditional ETL/PySpark).
- Concept: DBT CLI vs DBT Cloud.
- Hands-on: Install DBT CLI, sign up for DBT Cloud.
- Hands-on: Configure Snowflake connection using profiles.yml.
- Outcome: First dbt run connected to warehouse.

Session 2 – Understanding DBT Project Structure

- Concept: Anatomy of a project – models/, tests/, seeds/, snapshots/, macros/.
- Concept: dbt_project.yml and its role.
- Hands-on: Create a new project in CLI and DBT Cloud.
- Hands-on: Organize project folders for staging and marts.
- Outcome: Functional project skeleton with proper structure.

2. Core DBT Concepts (Sessions 3–8, 12 Hours)

Session 3 – Building Models

- Concept: Using ref() to define dependencies.
- Concept: Materializations – table, view, incremental, ephemeral.
- Hands-on: Create staging model with ref().
- Hands-on: Build fact_sales as incremental model.



- Hands-on: Building SQL transformations using ref().

Session 4 – Sources

- Concept: Declaring upstream data as sources.
- Concept: Testing and freshness checks.
- Hands-on: Configure sources in YAML.
- Hands-on: Run freshness check for raw tables.

Session 5 – Seeds

- Concept: Role of seeds for reference data.
- Hands-on: Add a CSV seed (product lookup).
- Hands-on: Query and join seeded data with models.

Session 6 – Testing Basics

- Concept: Importance of testing in pipelines.
- Concept: Built-in tests (unique, not_null, relationships).
- Hands-on: Apply built-in tests to customer/order tables.
- Hands-on: Write and run a simple custom test.

Session 7 – Snapshots

- Concept: Why snapshots (historical data tracking, SCD Type 2).
- Hands-on: Create snapshot configuration.
- Hands-on: Track changes in customer dimension.

Session 8 – Practice Lab on Core Concepts

- Hands-on consolidation lab: Create sources, models, seeds, apply tests, implement a snapshot.
- Outcome: End-to-end small pipeline using all core features.



3. DBT Workflow (Sessions 9–11, 6 Hours)

Session 9 – Development Workflow

- Concept: dbt run, dbt test, dbt build – when to use each.
- Hands-on: Execute the run-test-build cycle.

Session 10 – Project Organization

- Concept: Layered approach – staging, intermediate, marts.
- Hands-on: Refactor existing models into layers.

Session 11 – Debugging and Configurations

- Concept: Using dbt debug.
- Concept: Variables and configs in dbt.
- Hands-on: Debug broken model.
- Hands-on: Implement variable to control schema.

4. Advanced DBT Features (Sessions 12–15, 8 Hours)

Session 12 – Macros & Jinja

- Concept: Introduction to Jinja templating.
- Concept: Writing reusable macros.
- Concept: Parameterization.
- Hands-on: Parameterize model logic with Jinja.
- Hands-on: Create parameterized macro for standard joins.

Session 13 – Hooks & Operations

- Concept: Pre- and post-model hooks.
- Concept: dbt run-operation.
- Hands-on: Implement a post-hook for auditing.
- Hands-on: Run an operation to refresh schema.



Session 14 – Documentation & Lineage

- Concept: Documenting models.
- Concept: Generating lineage graphs.
- Hands-on: Add documentation to models.
- Hands-on: Generate docs site and explore lineage.

Session 15 – Environment Management

- Concept: Configuring dev/test/prod.
- Hands-on: Create multiple environments in DBT Cloud.
- Hands-on: Demonstrate schema separation.

5. Orchestration & Deployment (Sessions 16–18, 6 Hours)

Session 16 – Running DBT Locally vs Cloud

- Concept: Differences between CLI and Cloud runs.
- Hands-on: Run project in both CLI and Cloud.

Session 17 – Scheduling Jobs in DBT Cloud

- Concept: Job scheduling basics.
- Hands-on: Create and schedule job for daily staging refresh.

Session 18 – Orchestration Integrations

- Concept: Options beyond DBT Cloud (Airflow, Prefect).
- Hands-on (light): Demonstration of triggering DBT from Airflow.
- Hands-on: Monitor jobs and set up notifications in Cloud.

6. Review and Capstone (Sessions 19–20, 4 Hours)

Session 19 – Review & Troubleshooting Lab



- Hands-on: Debug broken model chains, test failures.
- Hands-on: Apply fixes and rerun project.

Session 20 – Evaluation Lab (2.5 Hours) + 1.5 Hrs of MCQ + 1 on 1 Viva

Learners build a DBT project from scratch, covering major topics from:

- Setup a new project.
- Add sources, models, seeds.
- Apply tests and snapshots.
- Add macros and documentation.
- Generate lineage site.
- Deploy in DBT Cloud and schedule a job.
- Present working pipeline at the end.